

# Ramdular Yadav

ML Engineer | GCP Certified  
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Github | LinkedIn | Portfolio

## Education

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### Tribhuvan University — Purwanchal Campus

2021 – 2025

*Bachelor of Computer Engineering*

- Equivalent GPA: 2.8/4.0 | First Division (65.87%)
- Thesis: AI-Based Online Exam Proctoring System

## Work Experience

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### Machine Learning Engineer

Jun 2026 – Sep 2026

*Entain (Client Engagement via Fusemachines Inc.)*

*Remote*

- Built a fraud and abuse detection system for **Entain**, a global online gaming and sports-betting platform, to identify suspicious players and coordinated abuser rings.
- Built an end-to-end Player 360 ML and Graph Analytics platform for abuser and multi-accounting detection across a 20M+ player ecosystem, supporting proactive fraud investigation and account-risk prioritization.
- Designed scalable BigQuery-based data pipelines integrating transactional, behavioral, and graph data across nodes such as players, devices, IP addresses, bonuses, and affiliates.
- Engineered behavioral and abuse-detection features covering deposits, withdrawals, bet flow, session activity, and timing-velocity, along with graph-based features such as connectivity, neighborhood statistics, and shared-entity patterns, using relationship and hub-size filtering to reduce graph noise.
- Developed XGBoost and Heterogeneous GNN models for abuse and multi-accounting detection, with the GNN achieving 0.715 PR-AUC and 0.95+ ROC-AUC on a highly imbalanced 4.6M+ player dataset; applied SHAP-based explainability to support analyst investigations.
- Built leakage-safe, time-aware training pipelines and a scheduled daily inference system with automated resource monitoring for production deployment.

### Machine Learning Engineer (Full-Time)

Sep 2025 – May 2026

*Fusemachines Inc.*

- Built an AI-powered platform that converts e-commerce product reviews into structured insights and sentiment summaries.
- Developed a system to analyze customer feedback and generate product improvement and design recommendations.
- Created automated workflows to classify reviews and highlight key issues, trends, and customer needs.
- Produced visual and text-based outputs to support better product decision-making.
- Designed and deployed a demand forecasting pipeline to optimize inventory planning for a furniture retail brand, reducing stockouts and overstock risk.
- Built and compared multiple time series and ML models (Linear Regression, XGBoost, Prophet), selecting the best-performing model based on MAE and RMSE.
- Engineered features including seasonality trends, lag variables, and rolling statistics to improve forecast accuracy.
- Implemented the Prophet model for seasonality-aware forecasting and benchmarked it against baseline ML approaches.

### AI/ML Intern

May 2025 – Sep 2025

*Fusemachines Inc.*

- Built an LLM-powered Interview Agent using LangChain for dynamic, multi-agent interview simulations.
- Designed and curated technical learning materials for the AI Fellowship program covering evaluation metrics, model stability, and self-supervised learning.

### AI Fellowship

Apr 2024 – Nov 2024

*Fusemachines Inc.*

- Completed a competitive 6-month fellowship focused on Machine Learning, Deep Learning, Computer Vision, NLP, and MLOps.
- Developed and deployed end-to-end AI solutions including predictive models, web scraping pipelines, and real-time applications using Streamlit and Docker.
- Explored advanced concepts including Transformers, Large Language Models, and Generative AI in research-driven and production-ready projects.

### Data Analyst

Nov 2023 – May 2024

*Karma Technology*

- Created and curated high-quality datasets for the team's data-driven projects, contributing to effective decision-making.

# Research Projects

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## AI-Based Online Exam Proctoring System

*Apr 2024 – Mar 2025*

- Developed a real-time AI-powered proctoring system; fine-tuned YOLOv8 to detect banned objects from webcam input.
- Used OpenCV and MediaPipe to track face, eye gaze, and head direction; implemented audio monitoring for unusual sounds.
- Detected tab switching and full-screen exits; generated automated behavioural reports every 5 minutes.
- Built with Django backend, achieving 90% accuracy in detecting cheating behaviours.

## Interview Agent

*Jun 2025 – Sep 2025*

- Built a multi-agent mock interview system using LangChain and Gemini LLM to simulate intelligent, job-specific interviews.
- Integrated custom *ResumeParserTool* and *JDParseTool* LangChain tools, allowing LLMs to dynamically reason over parsed resumes and job descriptions.
- Designed agents for Q&A, Evaluation, Follow-Up, and Chitchat to create personalized, human-like interview experiences.

## RAG-Based Advanced Multimodal Research Assistant

*Jun 2025 – Jul 2025*

- Developed an intelligent Q&A system for multi-format documents (PDF, Word, TXT) and images (charts, scanned notes).
- Enabled context-aware retrieval using ChromaDB for persistent vector storage of text and image embeddings.
- Implemented image captioning with Gemini Vision, semantic text chunking, and a modular architecture for scalability.
- Built a Streamlit UI and integrated RAGAS for response evaluation and hallucination reduction.

## Fake News Detection Using NLP and Deep Learning

*Feb 2025 – Jul 2025*

- Developed a classification pipeline using ML, DL (LSTM, CNN), and Transformers on WELFake and FakeNewsNet datasets.
- Achieved 98.2% accuracy on WELFake using fine-tuned BERT; 73.5% generalization accuracy using a BERT + CNN + LSTM hybrid on FakeNewsNet.
- Deployed a real-time prediction interface with Streamlit and containerized the solution using Docker.

# Technical Skills

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**Programming Languages:** Python, C/C++, SQL, Bash

**AI & Agentic Frameworks:** LangChain, RAG Systems, Prompt Engineering, Generative AI

**Machine Learning & Data Science:** Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Time Series Forecasting, XGBoost

**Deep Learning & Graph Analytics:** TensorFlow, Keras, PyTorch, PyTorch Geometric, Heterogeneous GNNs, OpenCV

**Natural Language Processing:** HuggingFace Transformers, SpaCy, NLTK, BERT, GPT, NER, Summarization

**Web & API Development:** FastAPI, Flask, Django, RESTful APIs, Streamlit

**Databases:** MySQL, PostgreSQL, Google BigQuery

**Cloud & Deployment:** Google Cloud Platform (GCP), Docker, Model Deployment

**Explainability & Evaluation:** SHAP, PR-AUC Optimization, RAGAS

**Workflow Automation:** n8n, Claude Code

# Certifications

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- Google Cloud Certified Professional Machine Learning Engineer (GCP)
- Machine Learning Nanodegree